Jialuo Du

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Research Interests

Low-power wide-area IoT, wireless physical layer, and AI for wireless systems. My research builds **physics-informed intelligent signal processing**, which joins white-box physical models with black-box learning: the models provide robustness and the learning provides adaptiveness, so that weak and distorted signals can be recovered on commodity IoT devices.

Professional Appointments

Aug 2025 – Present **Michigan State University**, *Research Associate (Postdoc)*

- Department of Computer Science and Engineering. Supervisor: Prof. Zhichao Cao

Education

Sept 2020 – July 2025 **Tsinghua University**, *Ph.D., School of Software*

- Advisor: Prof. Yunhao Liu

Sept 2016 – June 2020 **Tsinghua University**, *B.Eng., Computer Science and Technology*

Publications

Conference Papers

DeRa: Deep-Range Low-Power Wide Area IoT via Ultra-sensitive Signal Detection
Jialuo Du, Jingkai Lin, Zhichao Cao
ACM MobiCom 2026, pp. 599–619. Acceptance rate: 89/625 (14.2%). DOI

LoRaTrimmer: Optimal Energy Condensation with Chirp Trimming for LoRa Weak Signal Decoding
Jialuo Du, Yunhao Liu, Yidong Ren, Li Liu, Zhichao Cao
ACM MobiCom 2024. Acceptance rate: 103/494. DOI

SRLoRa: Neural-enhanced LoRa Weak Signal Decoding with Multi-gateway Super Resolution
Jialuo Du, Yidong Ren, Zhui Zhu, Chenning Li, Zhichao Cao, Qiang Ma, Yunhao Liu
ACM MobiHoc 2023. Acceptance rate: 30/136. DOI

NELoRa-Bench: A Benchmark for Neural-enhanced LoRa Demodulation
Jialuo Du, Yidong Ren, Mi Zhang, Yunhao Liu, Zhichao Cao
ICLR 2023 Workshop on Machine Learning for IoT. arXiv

SRPeek: Super Resolution Enabled Screen Peeking via COTS Smartphone
Jialuo Du, Chenning Li, Zhenge Guo, Zhichao Cao
IEEE ICPADS 2021. DOI

Kairos: Soil-Aware Transmission Control for Long-Term Agricultural Cross-Soil LoRa
Jingkai Lin, Yidong Ren, **Jialuo Du**, Yimeng Liu, Yawen Wang, Younsuk Dong, Tianxing Li, Zhichao Cao
IEEE ICNP 2026. Acceptance rate: 67/277 (24.2%). PDF

LoRaSeek: Boosting Denoising Ability in Neural-enhanced LoRa Decoder via Hierarchical Feature Extraction
Khang Nguyen, Yidong Ren, **Jialuo Du**, Jingkai Lin, Maolin Gan, Shigang Chen, Mi Zhang, Chunyi Peng, Zhichao Cao
ACM MobiCom 2025. Acceptance rate: 36/348. DOI

Demeter: Reliable Cross-soil LPWAN with Low-cost Signal Polarization Alignment
Yidong Ren, Wei Sun, **Jialuo Du**, Huaili Zeng, Younsuk Dong, Mi Zhang, Shigang Chen, Yunhao Liu, Tianxing Li, Zhichao Cao
ACM MobiCom 2024. Acceptance rate: 103/494. DOI

Prism: High-throughput LoRa Backscatter with Non-linear Chirps
Yidong Ren, Puyu Cai, Jinyan Jiang, **Jialuo Du**, Zhichao Cao
IEEE INFOCOM 2023. Acceptance rate: 252/1312. DOI

EyeLoc: Smartphone Vision-enabled Plug-n-play Indoor Localization in Large Shopping Malls
Zhichao Cao, **Jialuo Du**, Manni Liu, Qing Zhou
IEEE DySPAN 2019.

Research Agenda

Scaling LoRa to Massive IoT: A Physical Layer Approach
Jialuo Du
ACM MobiSys 2026 Rising Stars Forum (MobiSys Companion '26), pp. 150–151. DOI

Journal Articles

Umami-MRNN: Deep Learning-based Prediction of Umami Peptide Using RNN and MLP
Lulu Qi, **Jialuo Du**, Yue Sun, Yongzhao Xiong, Xinyao Zhao, Daodong Pan, Yueru Zhi, Yali Dang, Xinchang Gao
Food Chemistry, 2023. DOI

EyeLoc: Smartphone Vision-enabled Plug-n-play Indoor Localization in Large Shopping Malls
Manni Liu, **Jialuo Du**, Qing Zhou, Zhichao Cao, Yunhao Liu
IEEE Internet of Things Journal, 2020. DOI

Preprints

AIoT-based Continuous, Contextualized, and Explainable Driving Assessment for Older Adults
Yimeng Liu, Fangwei Zhang, Maolin Gan, **Jialuo Du**, Jingkai Lin, Yawen Wang, Fei Sun, Honglei Chen, Linda Hill, Ruofeng Liu, Tianxing Li, Zhichao Cao
arXiv:2603.00691, 2026. arXiv

NELoRa-Bench: A Benchmark for Neural-enhanced LoRa Demodulation
Jialuo Du, Yidong Ren, Mi Zhang, Yunhao Liu, Zhichao Cao
arXiv:2305.01573, 2023. arXiv

Research Highlights

Ultra-sensitive packet detection (DeRa, MobiCom 2026, first author)

- Showed that weak LoRa packets are lost at *detection*, before decoding starts: carrier frequency offset rotates the phase of every preamble chirp, so existing detectors can only add magnitudes.
- Modeled the phase evolution, compensated CFO-induced timing, frequency and leakage distortions (up to 40 kHz), and accumulated the preamble coherently, lowering the detection threshold by 4.3–4.8 dB and the decoding threshold by 3.3–3.9 dB on COTS devices across SF 7–12.

Optimal weak-signal demodulation (LoRaTrimmer, MobiCom 2024, first author)

- Found an overlooked frequency and phase jump inside every real LoRa symbol that standard dechirp-and-FFT demodulation ignores, and derived the optimal decoder under a probabilistic channel model.
- Computed all trimmed FFTs at once in $\mathcal{O}(n^{1.5})$ time by splitting the DFT matrix along its anti-diagonal. Gains 1.70–2.49 dB SNR over the state of the art, including the neural decoder NELoRa, at a fraction of its computation.

Neural and multi-gateway decoding (SRLoRa, MobiHoc 2023; NELoRa-Bench, ICLR-W 2023; first author)

- Designed a neural architecture with max-min-merge layers that fuses weak copies of one packet from multiple gateways.
- Released NELoRa-Bench, an open dataset of 27,329 real LoRa symbols and an evaluation methodology adopted by later neural decoders, including LoRaSeek (MobiCom 2025).

Open-Source Software and Datasets

- LoRaTrimmer decoder: github.com/LoRaTrimmer/LoRaTrimmer, with a reproducible workflow on Kaggle
- NELoRa-Bench dataset: github.com/daibiaoxuwu/NeLoRa_Dataset and Kaggle
- Browser-based interactive LoRa receiver demos: daibiaoxuwu.github.io

Teaching Experience

- Assisted in teaching CSE 410 Operating Systems (undergraduate), Michigan State University, Spring 2026

Mentoring

- Jingkai Lin, Ph.D. student, Michigan State University. Outcomes: first author of Kairos (IEEE ICNP 2026); second author of DeRa (ACM MobiCom 2026)

Professional Service

- Technical Program Committee member, IEEE International Conference on Mobility, Sensing and Networking (MSN), 2026
- Distinguished Reviewer Board, ACM Transactions on Internet of Things (TIOT), 2026